Sprint 3: Capstone Design

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Gates Arnold

Jackson Burke

Thomas Hicks

Cole Munn

Repository: https://github.com/jmburke4/capstone-design-manager
Website: https://jmburke4.github.io/capstone-design-manager/

Project Focus Reminder

APP SCOPE

- Accessible via a computer through the web
- Manages student projects
- Provides sponsors with feedback route
- Facilitates coordination between all users

SPRINT SUMMARY

- Minimal viable product
- Documentation
- Testing features
- Hardened deployment

WHAT REMAINS

- Functionality backlog
- Integration with My-Bama SSO
- Deployment configuration for Auth0
- Transfer project ownership

Prioritized Sprint Goals

01 *Account*

- Complete Sponsor workflow
- Complete Student Workflow
- Implement seamless Oauth

02 *Deployment*

- Deployment pipeline
- Secure HTTPS hosting
- Admin cloud access
- Deploy final version

03 *Documentation*

- Unit tests
- User docs
- Deployment docs
- Cloud docs
- Owner transfer docs

Major Sprint Achievements: Backend

- Limited file attachment functionality
- Unit Testing
  - Views
  - Emails
- Administrator tools
  - Viewing student preferences
  - Assigning and updating projects

Search 15 results (45 total)

Action: Assign to project Go 0 of 15 selected

STUDENT	PROJECT	RANK	SEMESTER	IS ASSIGNED	ASSIGNED
Jackson Burke	Enterprise Resource Planning System	1	Spring 2026	False	0
Jackson Burke	Cloud Data Analytics Dashboard	1	Spring 2026	False	0
Jackson Burke	Computer Vision Quality Control	1	Spring 2026	False	0
Jackson Burke	DevOps Automation Pipeline	1	Spring 2026	False	0
Jackson Burke	Distributed Systems Scheduler	1	Spring 2026	False	0
Jackson Burke	Edge Computing Infrastructure	2	Spring 2026	False	0
Jackson Burke	Graduate	2	Spring 2026	False	0
Jackson Burke	A Funky Space Reincarnation	2	Spring 2026	False	0
Jackson Burke	Blockchain Supply Chain Tracker	2	Spring 2026	False	0
Jackson Burke	I Got You Open	2	Spring 2026	False	0
Jackson Burke	Microservices Architecture Refactor	3	Spring 2026	False	0
Jackson Burke	Mobile Health Monitoring System	3	Spring 2026	False	0
Jackson Burke	Project Names are Overrated	3	Spring 2026	False	0
Jackson Burke	Quantum Computing Framework	3	Spring 2026	False	0
Jackson Burke	Real-time Collaboration Suite	3	Spring 2026	False	0

FILTER

Show counts

Clear all filters

↓ By semester

All
Fall 2026
Spring 2027
Spring 2026
-

↓ By Student Assigned

All
Assigned
Not Assigned

↓ By rank

All
1
2
3

↓ By Project

All
A Funky Space Reincarnation (0)
Blockchain Supply Chain Tracker (0)
Cloud Data Analytics Dashboard (0)
Computer Vision Quality Control (0)

Major Sprint Achievements: Backend

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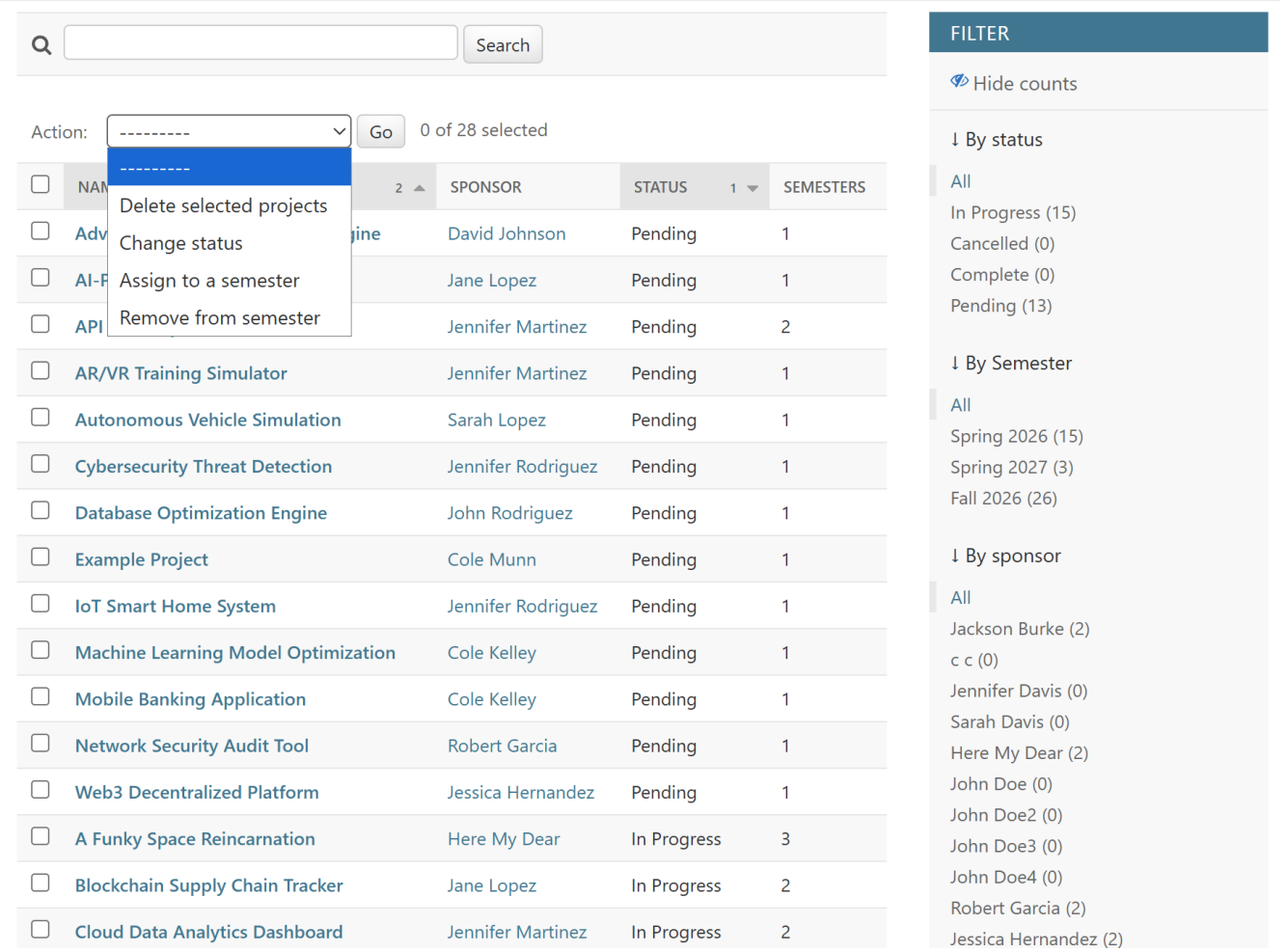
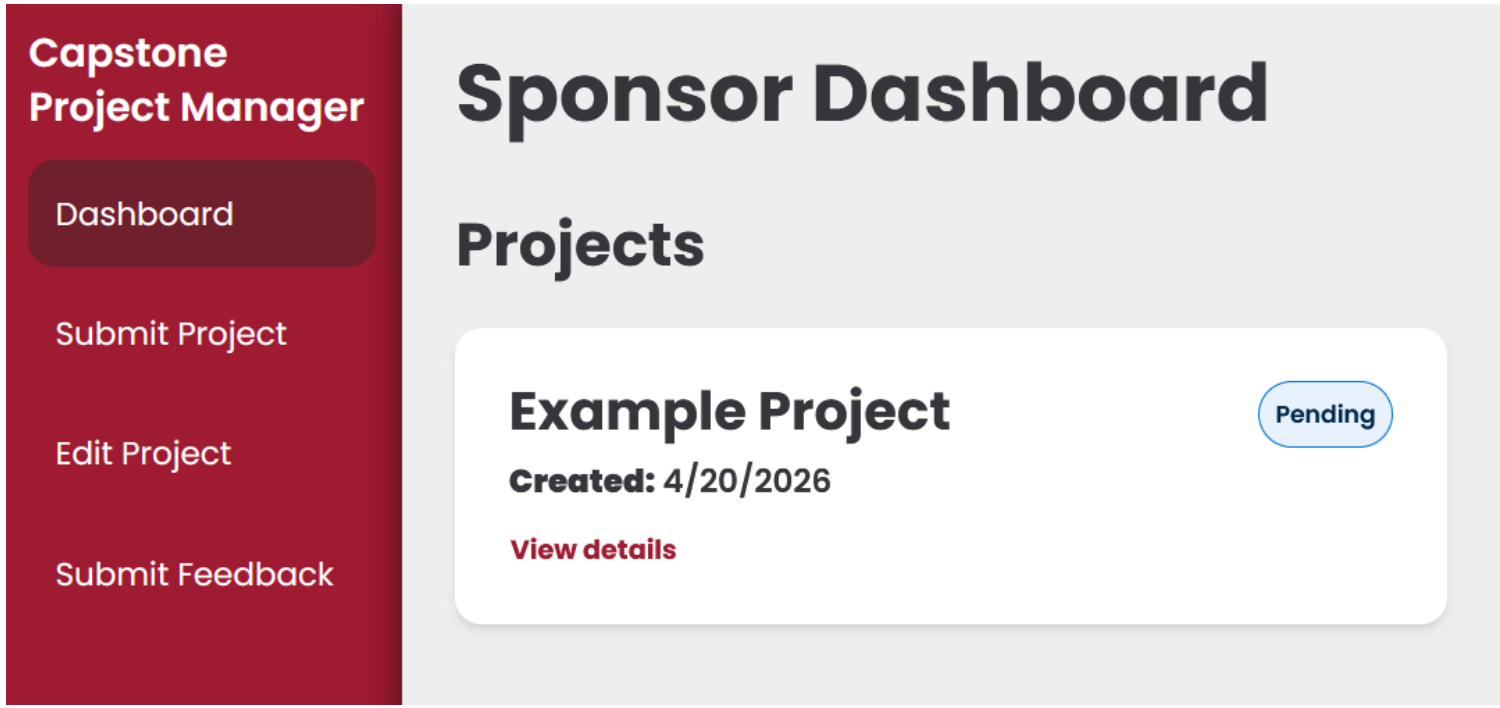


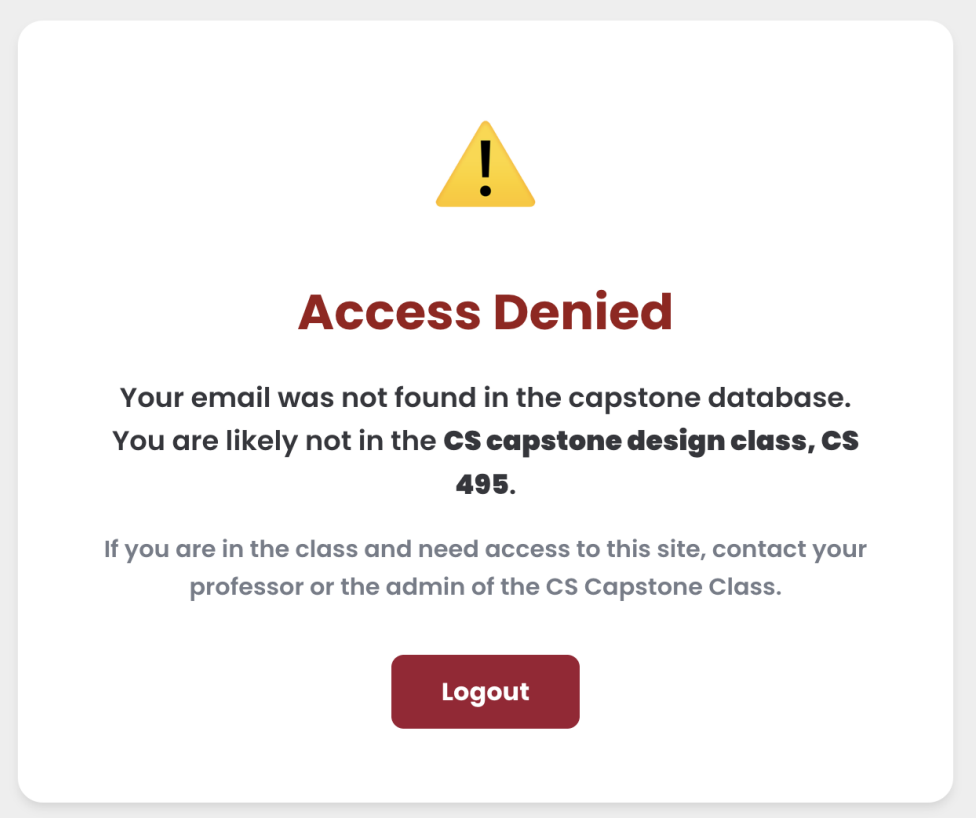
Major Sprint Achievements: Frontend

- Sponsor Workflow
  - Dashboard to track all existing projects
  - Project submission / editing
  - Feedback submission
- Student Workflow
  - Project Assignments
  - Data caching for session



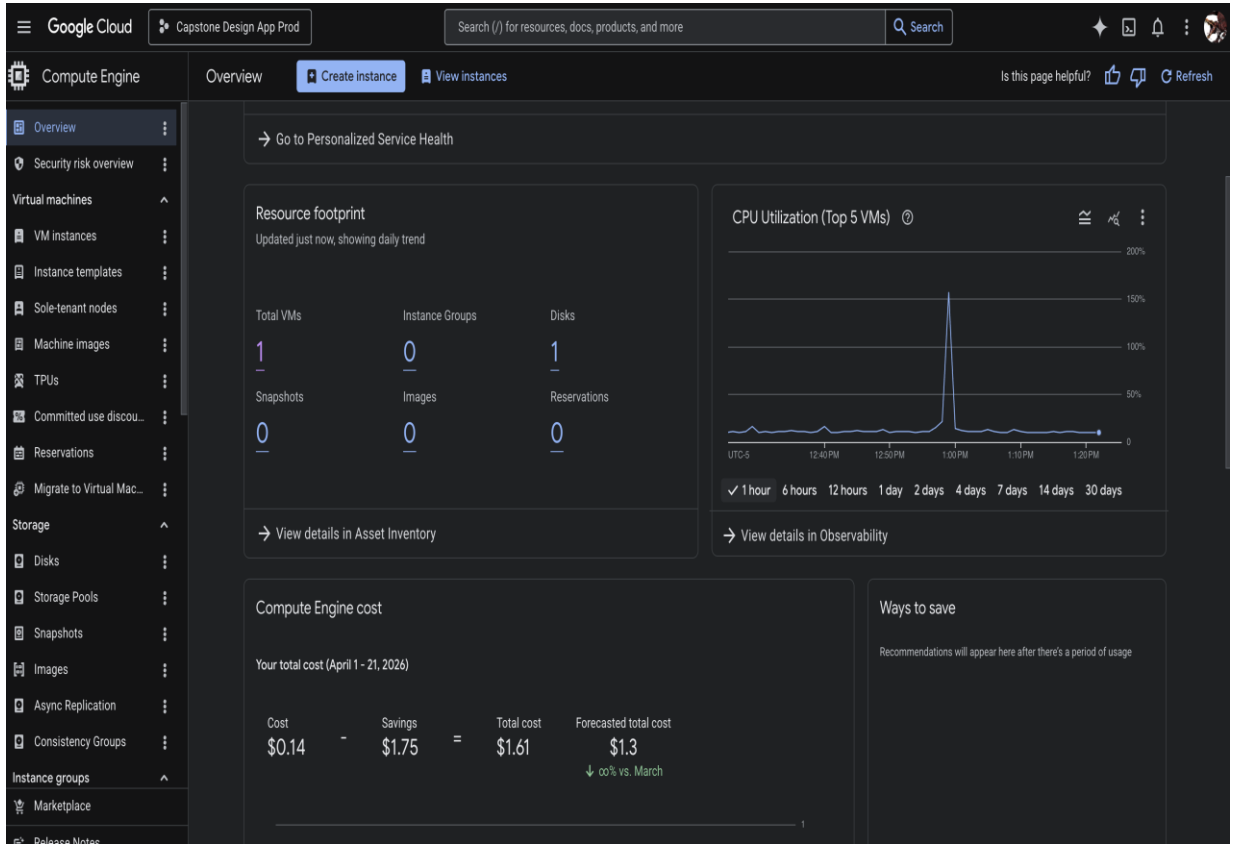
Major Sprint Achievements: OAuth + Cloud

- Auth0
  - Student + Sponsor roles
  - Admin role
  - Email verification
  - Better authentication
- Synopsis
  - Previously, account creation was messy and fragile
  - Now, lifetime roles are assigned at registration, and student accounts are better authenticated



Major Sprint Achievements: OAuth + Cloud

- Gcloud deployment
  - Custom https site
  - Cheap VM setup
  - Limited admin panel access
  - Semi-automated deployment
- Synopsis
  - Previously, there was no easy way to deploy https
  - Now, we have a deployment pipe-line which uses the Gcloud CLI

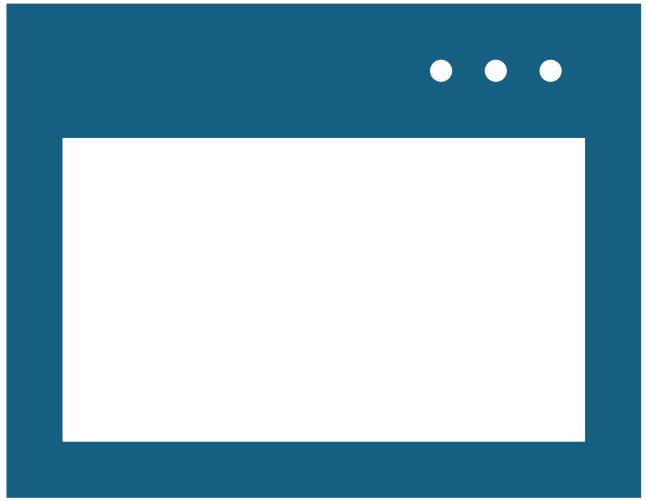


Unfinished Work for Next Group

- Project file attachments
  - Needs to be implemented on frontend for sponsor use
- Sponsor workflow
  - Display student contact info for assigned projects
- Finish caching storage for both Sponsor/Student workflows
  - Reduce amount of API calls per page
- Dynamic rank form requirements (currently requires at least 15 projects)

Project/Sprint Backlog

- Submission Deadline functionality (Student)
- Semester Model
  - Allows admins to select what projects are available to students
- Multi-project status update functionality (Admin)
- Required Authentication for DRF endpoints.



Cyber Security

Authentication

- Email verification
- Restrict user roles based on email domain Admin panel privileges

API Securement

- Each endpoint now requires a credentials token

Logs

Storage of log data of both actions taken in the dashboard by the administrators, as well as authentications made by your users. Show more ›

Ex: type:"s" AND date:[2019-05-22 TO *] | Filter: All | All Dates | Live Mode

To perform your search, press `enter`. Need help querying? Check out our query syntax.

Timestamp	Type	Description	Connection	Application	Occurred
2026-04-21T16:58:27.720Z	Success Exchange	Authorization Code for Access Token	N/A	capstone-design-man...	an hour ago
2026-04-21T16:58:27.261Z	Success Login	Successful login	Username-Password-...	capstone-design-man...	an hour ago
2026-04-21T16:58:06.366Z	Success Logout	User successfully logged out	google-oauth2	capstone-design-man...	an hour ago
2026-04-21T16:57:56.437Z	Success Exchange	Authorization Code for Access Token	N/A	capstone-design-man...	an hour ago
2026-04-21T16:57:55.822Z	Success Login	Successful login	google-oauth2	capstone-design-man...	an hour ago
2026-04-21T16:57:54.294Z	Warning During Login	You are using Auth0 development keys which ...	google-oauth2	N/A	an hour ago
2026-04-21T16:57:48.319Z	Failed Silent Auth	Login required	N/A	capstone-design-man...	an hour ago
2026-04-21T16:41:01.293Z	Success Exchange	Authorization Code for Access Token	N/A	capstone-design-man...	an hour ago
2026-04-21T16:41:00.934Z	Success Login	Successful login	google-oauth2	capstone-design-man...	an hour ago
2026-04-21T16:40:59.682Z	Warning During Login	You are using Auth0 development keys which ...	google-oauth2	N/A	an hour ago
2026-04-21T16:40:56.047Z	Success Logout	User successfully logged out	Username-Password-...	capstone-design-man...	an hour ago
2026-04-21T16:37:23.666Z	Success Exchange	Authorization Code for Access Token	N/A	capstone-design-man...	an hour ago
2026-04-21T16:37:23.157Z	Success Login	Successful login	Username-Password-...	capstone-design-man...	an hour ago

Sponsor Feedback

Meeting Synopsis

- Walked through the Django admin panel and demonstrated the script-based email export function

Further thoughts

- More meetings with our sponsor would have been optimal, especially closer to the deadline
- Overall, Our sponsor feedback has been positive

Accessibility & Inclusive Design Features

01 *Sign Up*

- Straight forward account setup
- SSO option for sponsor account
- Proper domain filtering for Students

02 *Account Services*

- Streamlined workflows
- Easy project management for sponsors
- Simple preference submission for students
- Project viewing for both accounts

03 *Admin Access*

- Easy admin creation
- Email management features
- User management
- File attachments

Contributions

- Sponsor/Student Workflows: Thomas, Cole M, Gates
- Admin Workflow and Actions: Jackson, Gates
- OAuth: Cole K
- Interface Design: Cole M, Gates
- Cloud Deployment: Cole K
- Backend Unit Testing: Jackson
- Models and API Refactoring: Jackson